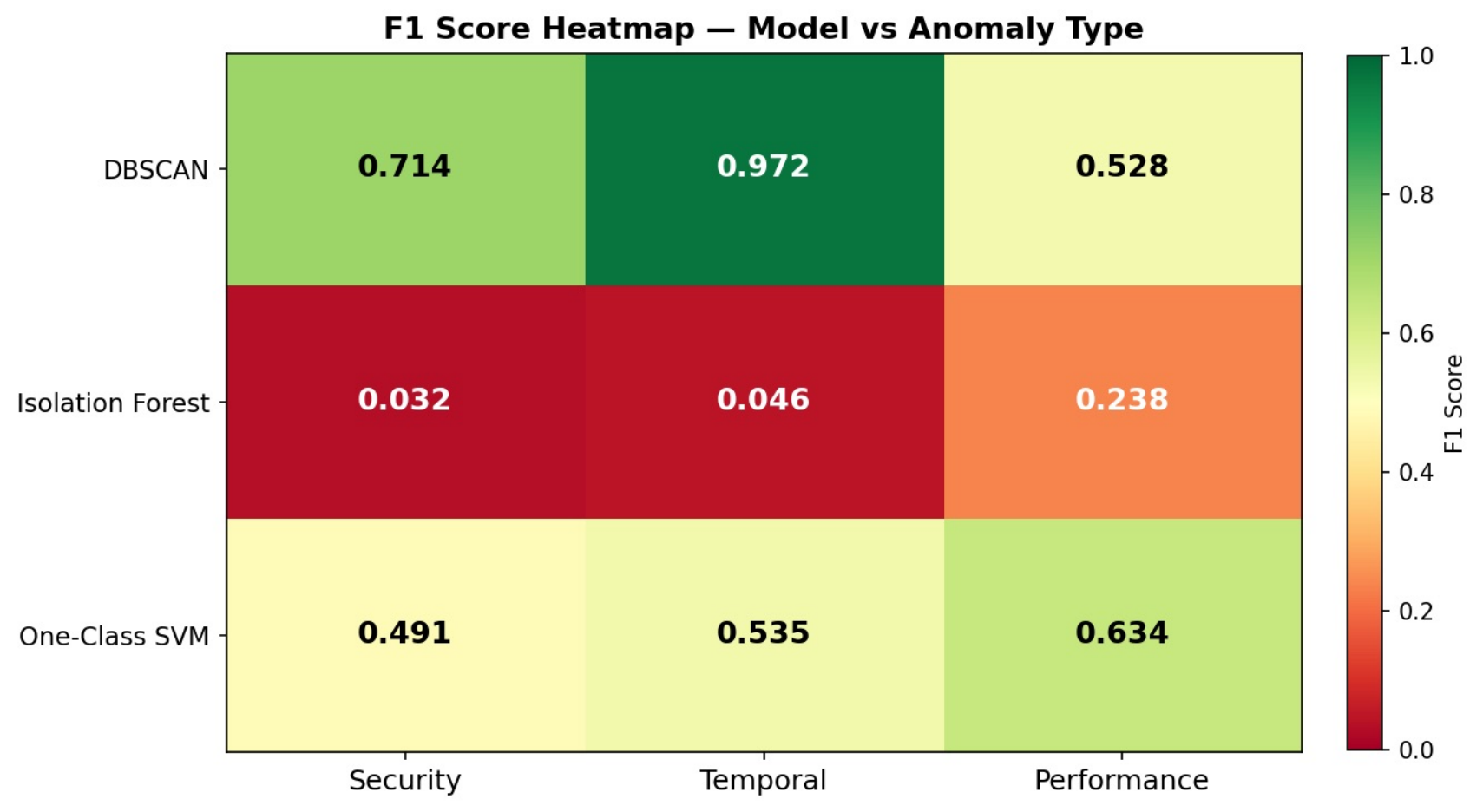
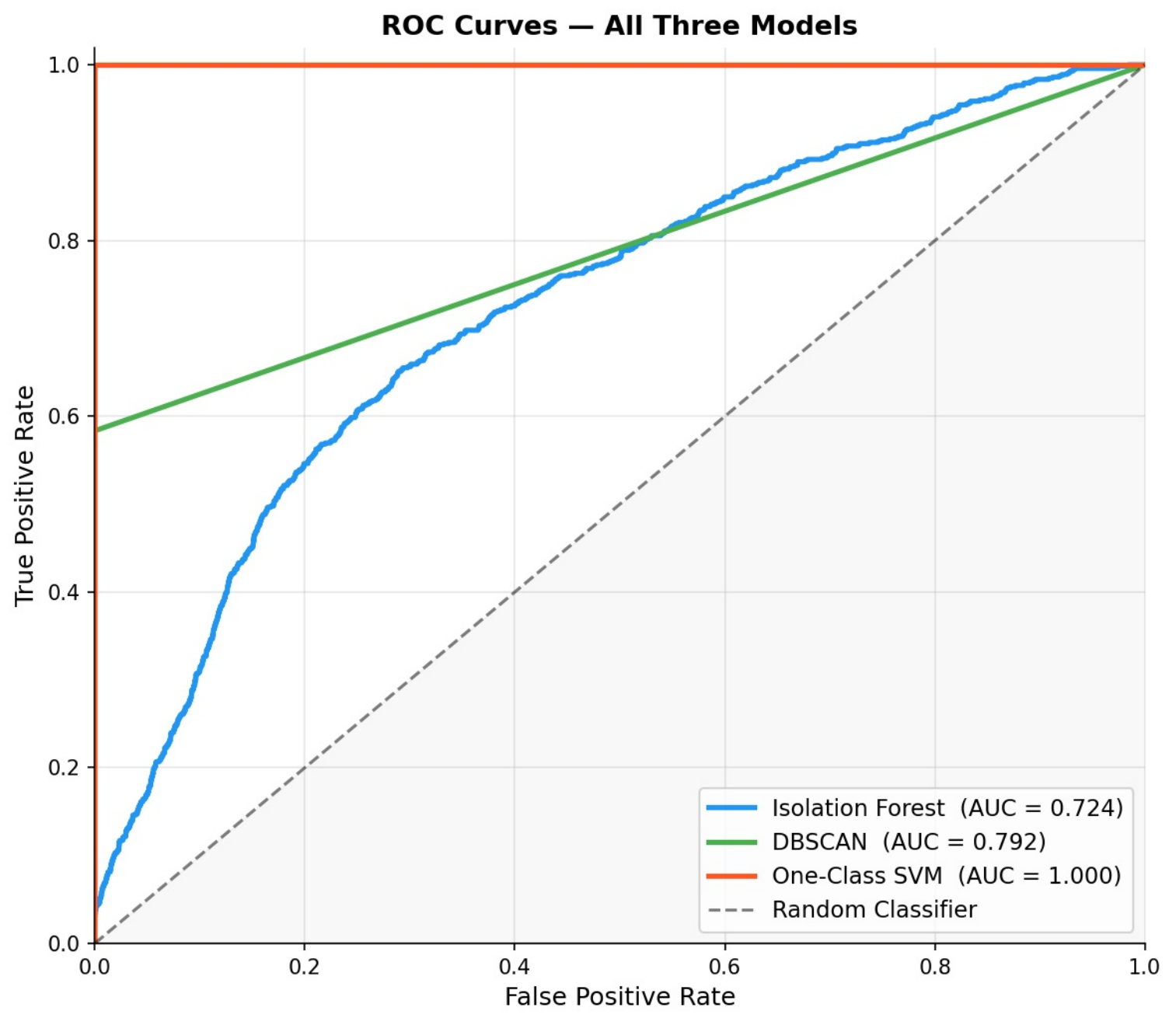
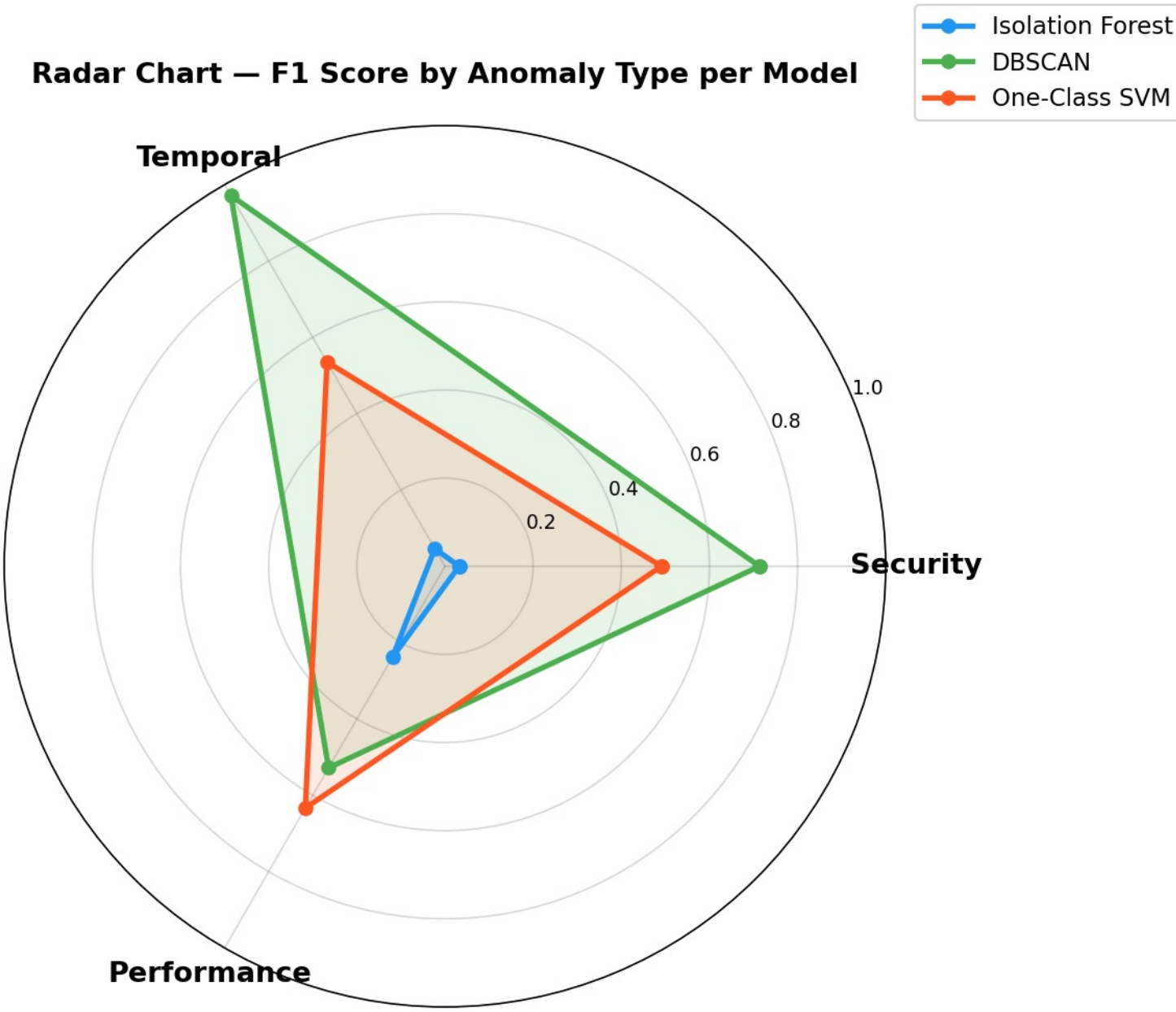
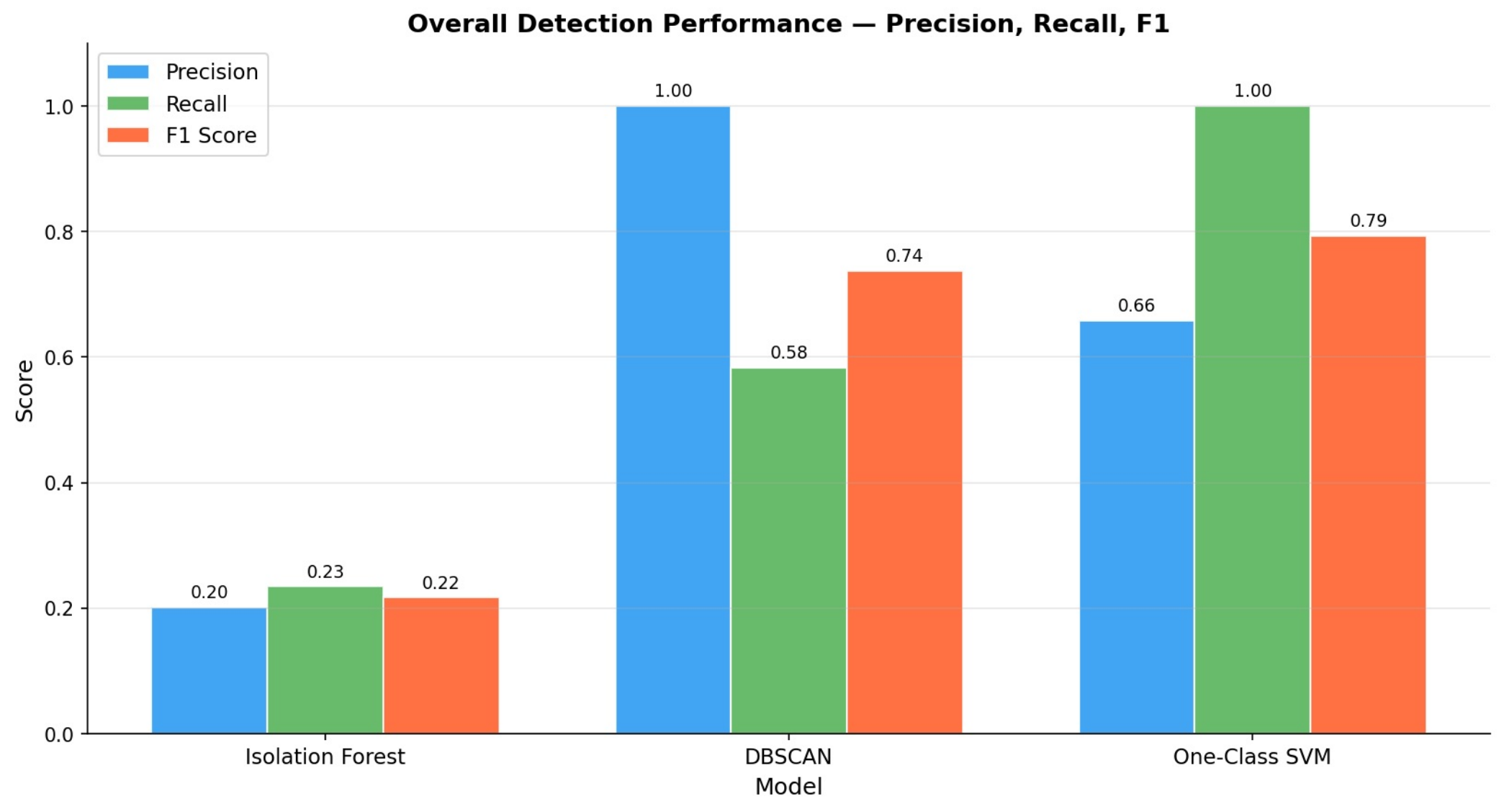
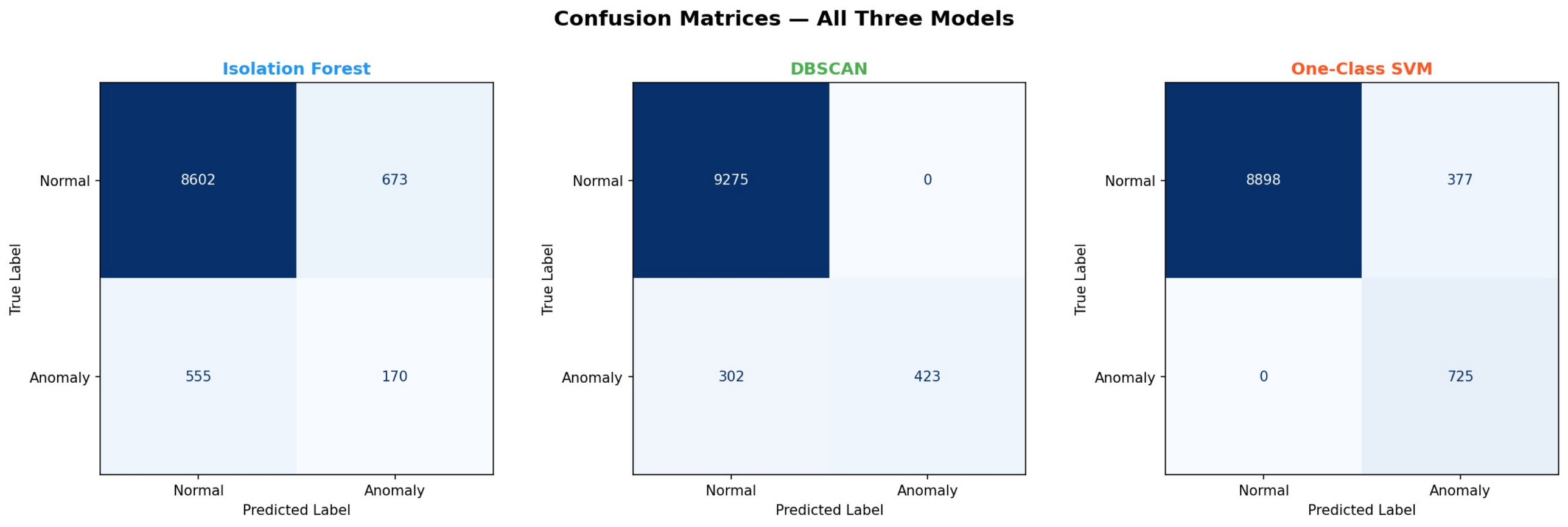

F1 Score Heatmap — Model vs Anomaly Type
DBSCAN
Isolation Forest
One-Class SVM
0.714
0.972
0.528
0.032
0.046
0.238
0.491
0.535
0.634
Security
Temporal
Performance
F1 Score
1.0
0.8
0.6
0.4
0.2
0.0


ROC Curves — All Three Models
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Isolation Forest (AUC = 0.724)
DBSCAN (AUC = 0.792)
One-Class SVM (AUC = 1.000)
Random Classifier


Radar Chart — F1 Score by Anomaly Type per Model
Isolation Forest
DBSCAN
One-Class SVM
Temporal
Security
Performance
0.2
0.4
0.6
0.8
1.0


Overall Detection Performance — Precision, Recall, F1
Precision
Recall
F1 Score
Score
1.0
0.8
0.6
0.4
0.2
0.0
0.20
0.23
0.22
1.00
0.58
0.74
0.66
1.00
0.79
Isolation Forest
DBSCAN
One-Class SVM
Model


Confusion Matrices — All Three Models
Isolation Forest
DBSCAN
One-Class SVM
Normal
Anomaly
True Label
8602
673
555
170
9275
0
302
423
8898
377
0
725
Normal
Anomaly
Predicted Label